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Digital transformation in emerging markets: Adoption dynamics of AI image generation in marketing practices

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ABSTRACT

This study explores the key variables influencing the adoption of AI image generators in marketing in Bangladesh, emphasizing the mediated impact of perceived usefulness. The study examines the present extent of AI image generator use across marketing agencies in Bangladesh and offers suggestions for optimizing their application to enhance efficiency, creativity, and competitiveness. A quantitative research strategy was used, gathering primary data via a standardized questionnaire administered to 480 marketing experts. Following the screening process, 320 replies were considered valid. Convenience sampling was used, focusing on professionals engaged in choices about the implementation of AI image generators. The data was analyzed using Smart PLS, a statistical program for SEM, to investigate the connections between key components and behavioral intents. Research demonstrates that PE, EE, SI, FC, and ATT substantially affect the motivation to use AI image generators. Nonetheless, PR exerts no substantial influence. PU mediators the associations between PE, SI, and PR with BI, but it does not mediate EE, FC, or ATT. The research underscores the critical importance of perceived usefulness to boosting adoption. This study offers actionable insights for marketing companies, highlighting the need to augment perceived usefulness by showcasing the advantages of AI. It enhances the comprehension of AI adoption in developing areas, facilitating digital transformation. This research is novel in its contextual emphasis on Bangladesh and its use of perceived usefulness as a mediating variable.

1. Introduction

In today's rapidly changing digital world, artificial intelligence (AI) has become an essential part of modern business operations, with its impact being especially noticeable in marketing. AI technologies are not only changing how companies understand consumer behavior but are also transforming how content is created and delivered [1–3]. Among the various AI-powered innovations, AI image generators have stood out as a disruptive force, allowing marketers to produce visually engaging and contextually relevant images with just a simple text prompt [4]. This advancement has made it possible to simplify creative workflows, reduce reliance on human graphic designers, and speed up content production across multiple digital platforms. As marketing increasingly relies on visual content and algorithm-driven engagement, the ability to create personalized, high-quality visuals has become a key competitive factor [5]. Already, AI-generated images are being utilized in social media campaigns, advertisements, email marketing, and e-commerce product displays, providing brands with a cost-effective solution to maintain visual consistency and capture consumer attention on a large scale. However, decisions about adopting such technology are not based solely on an organization's capacity or technical setup; rather, they are also shaped by individual perceptions about how relevant, useful, and easy the technology is to integrate [6,7]. This is particularly true in developing countries like Bangladesh, where digital transformation is still in progress and where individual cognitive and behavioral factors significantly influence adoption outcomes [8].

The integration of AI into creative and marketing fields has advanced rapidly, particularly with the rise of AI image generators that use sophisticated technologies such as Generative Adversarial Networks (GANs) and large language models [9,10]. These tools enable users to generate unique and context-appropriate visuals from text inputs, thereby changing the way marketing content is created and delivered. Globally, many brands have started to include AI-generated visuals in their content strategies to increase creative efficiency, reduce production time, and maintain brand uniformity across platforms [11,12]. This trend represents a larger shift toward automation and personalization in digital marketing, where AI is both a content facilitator and a strategic resource. While AI image generation has been quickly adopted in developed markets supported by advanced digital infrastructure and high technological readiness, the situation is different in emerging economies like Bangladesh, where adoption is more gradual and depends heavily on context. The digital economy in Bangladesh has been growing steadily, with increasing internet access, a young, tech-savvy population, and a rise in e-commerce and digital services [13]. Particularly, startups and SMEs have shown interest in AI tools to expand their marketing reach and enhance operational flexibility. Nevertheless, adoption decisions are often influenced by how usable technology is perceived to be, local market characteristics, cost concerns, and cultural

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fit [6]. Although the larger infrastructure continues to improve, the perceptions of individual marketers at the micro-level play a crucial role in shaping adoption patterns. In a resource-constrained yet digitally ambitious country like Bangladesh, how users view the usefulness and relevance of AI-driven tools is critical to understanding how these technologies become part of mainstream marketing. Thus, this study is positioned at the intersection of global technological progress and localized user behavior, providing a rich context for examining the dynamics of AI adoption in marketing.

Bangladesh has been undergoing a steady transformation toward digitalization, fueled by government initiatives, expanding internet penetration, and the growing digitization of various services [14]. While sectors such as banking, healthcare, and telecommunications have already begun applying AI-based solutions in customer service, diagnostics, and automation, marketing has started to show more openness to AI technologies. Digital marketing agencies, brand managers, and content creators have begun experimenting with AI for tasks like customer segmentation, campaign management, and content creation [3]. Among these, AI image generators tools that produce visuals from text prompts have become attractive options for marketers seeking to reduce design time and boost engagement through visuals. This technology is particularly appealing to SMEs in Bangladesh, many of which lack in-house creative teams or the budget to outsource graphic design work. These AI tools allow marketers to quickly create promotional materials, social media posts, and advertising visuals that align with their brand identity. However, the actual use of these technologies remains inconsistent across organizations. While some early adopters have incorporated AI-generated images into their workflows, many others still rely on traditional design methods, often due to a lack of technical skills, skepticism toward automation, or concerns about return on investment [10,15]. Moreover, the adoption process seems influenced by both individual and organizational readiness, including how useful marketers perceive these tools to be, how well the technology fits with current systems, and the availability of supportive conditions such as training or infrastructure [6,16]. As digital marketing grows in importance for Bangladeshi businesses, understanding the cognitive and environmental factors behind adoption behavior becomes even more relevant. Currently, the market is in a transitional phase where awareness and experimentation with AI image generators are on the rise, but full integration into marketing strategies is still evolving.

AI technologies have increasingly found their way into the marketing domain, bringing about advanced tools like AI image generators that enable marketers to create visually appealing content more efficiently and at scale [17]. Despite the promising potential of these technologies, their adoption especially in developing economies such as Bangladesh has been somewhat uneven and fragmented. While large global companies have quickly integrated AI image tools into their branding and creative workflows, marketers in emerging markets often approach these innovations more cautiously. This cautiousness seems to be influenced by a variety of behavioral, technological, and contextual factors, which have been noted but not fully explored in academic research [18]. Past studies on technology acceptance have highlighted constructs such as Performance Expectancy (PE) and Effort Expectancy (EE) as key predictors of technology use [19,20]. However, when it comes to new technologies in creative fields, personal and social factors such as Attitude Towards Technology (ATT), Social Influence (SI), and Perceived Risk (PR) also appear to play a significant role [21]. In Bangladesh, where digital infrastructure is gradually improving but technological exposure varies widely between industries, Facilitating Conditions (FC) can either promote or hinder the adoption process, particularly among small and medium enterprises. Although these variables have been studied individually, limited empirical work has been conducted to examine how they might interact together within a unified model to impact the Behavioral Intention (BI) to adopt AI-powered marketing tools. More specifically, the mediating effect of Perceived Usefulness (PU) a core construct of the TAM has not been adequately investigated in relation to AI image generators in the Bangladeshi context. Because of this, there is a restricted understanding of how user beliefs indirectly affect adoption through PU, which makes it difficult to explain why some marketers adopt these technologies more readily than others, even when they have similar access and exposure. Furthermore, earlier research has focused broadly on AI adoption in areas such as data analytics, automation, and customer service [22]. Only a few studies have paid attention to the creative and visual applications of AI in marketing, particularly from the viewpoint of end-users like marketing professionals. As the marketing landscape in Bangladesh becomes increasingly competitive and digitally driven, it becomes important and timely to understand how these behavioral and environmental factors together influence marketers' intentions to adopt AI image generators.

RQ1: To what extent are the BI of marketing professionals in Bangladesh influenced by factors such as PE, EE, SI, FC, PR, and their ATT when it comes to adopting AI image generators?

RQ2: In what way does PU act as a mediator between these factors PE, EE, SI, FC, PR, and ATT and the BI to adopt AI image generators?

The main aim of this study is to explore the behavioral factors that affect the intention of marketing professionals in Bangladesh when it comes to adopting AI image generators. Particular attention is given to the mediating role that perceived usefulness might play in this process. Several important constructs, including PE, EE, SI, FC, PR and ATT, are examined to see how they together influence the BI. It is hoped that through this investigation, a clearer picture can be formed of the various cognitive and contextual elements that either promote or hinder the use of AI-powered visual tools in marketing activities, especially within the rapidly digitalizing environment of Bangladesh. This understanding is considered valuable for both researchers and practitioners who are interested in the technological evolution of marketing in emerging economies.

The novelty of this study is primarily found in its focus on AI image generators, which remain an emerging and relatively underexplored area within the wider field of AI adoption in marketing. While much of the existing research has focused on AI applications such as customer analytics, chatbots, and recommendation systems, comparatively little empirical attention has been given to creative AI tools especially those designed to automate visual content creation [23]. In this regard, the study attempts to provide a focused examination of how marketing professionals in a developing country perceive and intend to adopt AI-generated images, which helps to address a notable gap in the current knowledge base. Furthermore, the research introduces a behavioral model that combines several individual, technological, and social factors including PE, EE, SI, FC, PR, and ATT. Within this framework, PU is uniquely positioned as a mediating variable. Although the mediating role of PU has been widely acknowledged in technology acceptance research [24], its specific effect on the adoption of creative AI tools in emerging markets like Bangladesh has not been fully explored and thus remains an area ripe for investigation. From a practical perspective, the study gains significance by capturing insights from Bangladesh a rapidly digitizing economy where the marketing sector is only beginning to experiment with AI-driven solutions. Because of resource limitations and diverse levels of digital readiness among firms, the findings are expected to offer valuable information for both practitioners and policymakers regarding factors that either enable or hinder AI image generator adoption. In the end, this study is hoped to contribute to theoretical progress in understanding technology adoption as well as provide practical guidance on the integration of AI tools into marketing strategies in similar socio-economic environments.

2. Literature review

The rise of AI has noticeably transformed traditional marketing approaches by bringing in innovative tools like AI image generators, chatbots, and data analytics platforms [3,4,10]. Among these technologies, AI image generators have been gaining increasing attention, mainly because they allow marketers to create visual content quickly and efficiently something that's become very important for digital marketing campaigns relying heavily on engaging and personalized images [25]. Despite this potential, however, empirical research looking into the behavioral factors that influence the adoption of AI image generators, especially in emerging markets, still appears to be quite limited. Various studies have examined the adoption of AI in marketing through different theoretical perspectives. For example, [26] applied the Unified Theory of Acceptance and Use of Technology (UTAUT) to study AI-powered chatbot adoption and found that factors like PE, SI, and FC played significant roles in shaping BI. Similarly, research by [27] on AI-driven data analytics adoption among SMEs in Bangladesh showed that EE and FC were key factors affecting acceptance. Though these studies provide important insights into AI adoption, most tend to focus on data-driven or interaction-focused AI applications, leaving creative AI tools like image generators less explored. The creative aspect of AI introduces some unique challenges and perceptions because users are not only expected to trust the usefulness of technology but also its ability to replicate or even enhance human creativity [28]. In this regard, PR becomes a noteworthy concern, as users may worry about the quality, originality, and ethical aspects of content generated by AI [29].

PU has consistently been recognized as a crucial factor influencing technology adoption in numerous studies [30,31]. Specifically, in the case of AI image generators, perceived usefulness reflects how much marketers believe these tools can improve their creative productivity and marketing effectiveness. Still, the mediating role of perceived usefulness between various behavioral factors such as PE, EE, SI, FC, attitude toward technology, and PR and BI has not been extensively studied, especially within the Bangladeshi marketing context. Additionally, Bangladesh offers a unique digital marketing environment characterized by rapid digital growth alongside certain infrastructural and resource challenges [32]. As interest in AI technologies grows among startups and SMEs, understanding the interplay of individual and organizational factors affecting adoption becomes even more important. This study builds existing research by integrating multiple behavioral, social, and technological factors within a single framework to better understand AI image generator adoption. Unlike previous works that focused mainly on general AI or specific applications, this research emphasizes the creative marketing tools dimension and highlights the mediating role of perceived usefulness, thereby adding fresh perspectives to AI adoption studies in developing countries.

This study has employed UTAUT as the core theoretical framework to understand BI towards adopting AI image generators in marketing strategies. The choice of UTAUT was made because it integrates components from eight major technology acceptance theories, thus providing a comprehensive yet straightforward model to explain users' intentions and actual use of technology [31,33]. Its robustness has been tested across a variety of technologies and cultural contexts, making it particularly suitable for studying new AI tools in a developing country such as Bangladesh. One of the main reasons for selecting UTAUT over other popular models like the TAM or the TPB is its inclusive nature. The UTAUT model does not just consider individual perceptions but also accounts for SI and FC alongside performance and effort expectancies [19,24]. This broader viewpoint is especially important when looking at AI image generators, where factors such as social norms, organizational backing, and infrastructure readiness often play crucial roles in determining whether marketers will adopt these tools. Additionally, UTAUT is known to be quite effective in explaining BI in scenarios involving novel or complex technologies. AI image generators represent a relatively new innovation that challenges traditional marketing methods. The theory's ability to capture various dimensions of influence including how users evaluate usefulness and ease of use, as well as external factors like peer pressure and available resources makes it a useful lens for analyzing adoption behaviors [19]. Since marketers in Bangladesh work in diverse organizational settings and face different levels of digital maturity, the model's inclusion of FC allows for a detailed examination of both enabling and limiting environmental factors, alongside personal attitudes. Several recent studies have applied UTAUT to explore AI adoption in marketing and related areas. For example, [26] used the UTAUT framework to investigate AI chatbot adoption in customer service and found that PE and SI were significant predictors of intention to adopt. Likewise, [27] applied UTAUT to study AI-driven data analytics adoption in Bangladeshi SMEs, emphasizing the importance of FC and EE in shaping user acceptance. These examples show the model's adaptability and explanatory strength across different contexts, supporting its use in this study. Moreover, UTAUT ability to predict behavior and its comprehensive nature align well with the focus on perceived usefulness as a mediating factor in this research. Although perceived usefulness is embedded within UTAUT, explicitly considering its mediating role helps to clarify how users' perceptions of utility influence the relationship between other factors and BI [31,34]. This detailed insight is necessary to understand the specific challenges and opportunities that come with AI image generators in marketing. A number of recent research studies have effectively used UTAUT to examine AI adoption in marketing and allied areas. For example, [26] utilized the UTAUT framework to investigate AI-based chatbot adoption in customer service, reporting that PE and SI had a notable impact on adoption intentions. In addition, [35] leveraged UTAUT to study AI-driven data analytics adoption in Bangladeshi SMEs, emphasizing the key roles of FC and EE in promoting user acceptance. Moreover, contemporary research has examined consumers' perceptions in metaverse commerce using an integrated UTAUT2 and dual-factor theory model, offering additional understanding of technology adoption dynamics in emerging digital environments [36]. These studies illustrate the model's flexibility and ability to explain adoption behavior in both global and local contexts, supporting its applicability for the present study.

The UTAUT framework provides a well-established, empirically supported, and context-sensitive foundation to explore technology adoption. Its consideration of individual, social, and organizational factors makes it the most appropriate theoretical model to study the adoption of AI image generators in Bangladesh's rapidly evolving marketing environment.

PE generally refers to how much an individual believes that using specific technology will help them perform their tasks better or improve their job performance [37]. When it comes to AI image generators, PE reflects how marketing professionals perceive these tools as helpful in boosting their creative productivity, enhancing the effectiveness of marketing campaigns, and ultimately contributing to improved performance results. Since marketing today depends so much on visually engaging content, being able to quickly produce high-quality images is often seen as a significant advantage [38]. Even though PE and PU are theoretically connected, they vary in focus and role. PE relates to job-specific anticipations about how technology will enhance effectiveness in specific work tasks, such as producing top-notch marketing visuals quickly. PU, conversely, represents a wider mental assessment of the technology's general usefulness, encompassing creative enhancement, competitive edge, and process integration [24]. In this study, PE acts as a precursor affecting PU, while PU bridges the influence of PE and other variables on BI, offering a deeper insight into adoption behavior in marketing contexts.

Many prior studies have found that PE is one of the most consistent and strong predictors of BI across various technologies and contexts. For example, [39] showed that PE had a significant influence on users' intentions to adopt AI-powered chatbots in customer service. Likewise, [40] found that PE was a key factor driving the acceptance of AI-based data analytics tools among SMEs in Bangladesh. These results support the idea that when people expect technology to improve their work performance, they are generally more willing to use it. In particular, for creative marketing fields where time and innovation are essential, the belief that AI image generators can speed up content creation without losing quality can be a strong motivator for adoption [41]. This is especially true in emerging economies like Bangladesh, where marketers often face resource limitations and intense competition, pushing them to adopt technologies that promise clear performance benefits. Furthermore, the rapid digital growth in Bangladesh encourages marketers to use AI tools that help them stay flexible and responsive to the fast-changing demands of consumers [13,42].

Therefore, based on substantial theoretical and empirical evidence, it seems justified to propose that a higher level of PE positively affects the BI to adopt AI image generators in marketing. This connection highlights how marketers value the expected practical benefits of AI-generated visuals in improving their professional effectiveness.

H1: PE positively influences BI to Use AI Image Generators.

EE basically refers to how easy someone thinks it is to use a particular technology [31,43]. When it comes to AI image generators, EE reflects the extent to which marketing professionals feel they can quickly learn and smoothly navigate these tools to create visual content without facing too many problems. Since these AI image generators are quite new and somewhat complex, the perception of how user-friendly they are becomes really important to encourage marketers to actually use them, especially for those who might not have strong technical skills [38]. It has been repeatedly shown through research that EE has a significant effect on the intention to adopt new technologies across different settings. For example, in a study by [27], EE was found to be a strong predictor for adopting AI-based data analytics among SMEs in Bangladesh, suggesting that when users find a technology easy to use, their willingness to adopt it tends to increase. In addition, [26] pointed out that ease of use had a positive influence on users' intentions to implement AI chatbots, highlighting that simpler technologies help reduce barriers to adoption and boost user confidence. In marketing, where professionals often work under tight deadlines and need to generate content quickly, tools that don't require much effort to learn or operate are naturally more attractive [13]. This is particularly true in developing countries like Bangladesh, where there can be large differences in digital literacy among users, and technologies seen as complicated or hard to use may be met with hesitation despite their advantages [38]. Therefore, perceiving AI image generators as easy to use can play a big role in shaping marketers' intentions to adopt these tools. Moreover, since the marketers involved in this study come from varied educational and technical backgrounds, ease of use becomes an even more critical factor affecting their readiness to incorporate AI tools into their routine work. Based on existing literature and practical realities, it seems reasonable to expect that higher EE will have a positive effect on the BI to use AI image generators.

H2: EE positively influences BI to Use AI Image Generators.

SI generally refers to how much people feel that important others like their colleagues, supervisors, peers, or industry leaders think they should use a certain technology [31,44]. When it comes to AI image generators, SI reflects the way social norms and outside pressures affect marketers' decisions about whether or not to adopt these new tools. Because adopting new and advanced technologies often comes with some uncertainty, marketers often rely on signals from their social circles to understand how acceptable and legitimate AI image generators are within their professional environment [38]. Previous studies have suggested that SI plays a pretty important role in shaping BI, especially in organizational and professional contexts. For example, [45] showed that SI had a strong positive effect on people's intention to use AI-powered chatbots, showing how peer recommendations and encouragement from managers can make a difference. Moreover, [18] found that SI shaped the intentions of SME managers in Bangladesh to adopt AI-driven analytics, highlighting how organizational culture and the opinions of peers matter a lot in emerging markets. In marketing specifically, where teamwork, creativity, and innovation often happen through social interaction, the views of trusted peers and industry experts can have a big impact on whether someone chooses to adopt new technology [46]. This is particularly true in Bangladesh, where marketers tend to work within close professional networks and are influenced by industry trends as well as what leading companies do [47]. Besides, SI can help reduce the PRs and doubts about new technology by giving social validation, which is important when people consider adopting something relatively new like AI image generators. Considering all these points, it is quite reasonable to expect that SI would have a positive effect on marketers' intention to adopt AI image generators. Understanding this connection better can help companies and technology providers create strategies that use social networks and opinion leaders to boost adoption rates.

H3: SI positively influences BI to use AI image generators.

FC basically refer to how much a person believes that the organizational and technical infrastructure is in place to support using a technology [31,48,49]. When it comes to AI image generators, FC includes things like having the right hardware, stable internet connection, training opportunities, and technical support that help marketing professionals actually use these AI tools in their daily work. Since AI image generators usually need some level of technological readiness, feeling that these FC exist plays an important role in whether someone intends to adopt the technology or not. Many previous studies have pointed out the importance of FC in adopting new technology. For example, [40] found that FC had a significant impact on the intention to adopt AI-based data analytics tools among SMEs in Bangladesh, showing how important infrastructure and support are. Similarly, [50] noted that FC positively influenced users' intentions to use AI chatbots, emphasizing that organizational backing and technical resources can reduce the barriers to adoption. In developing countries like Bangladesh, where there are still many infrastructural and tech challenges, FC become even more crucial. Marketing professionals might hesitate to use AI image generators if they feel the current technological environment doesn't really support smooth usage [38]. Considering the wide range of digital maturity across companies, especially among startups and SMEs, making sure that users feel supported through proper training, IT help, and compatible systems can boost their confidence and increase their willingness to adopt these AI marketing tools. Furthermore, FC also help lower perceived difficulty and risk by assuring users that assistance is available if technical problems come up. This kind of support encourages marketers to try out AI image generators more comfortably, which in turn leads to a higher intention to adopt. So, both theoretically and practically, it makes sense to expect a positive link between FC and BI to use AI image generators.

H4: FC positively influences BI to use AI image generators.

ATT basically reflects an individual's overall positive or negative feelings about using a specific technology [18,51,52]. When it comes to AI image generators, ATT is about how marketers view the idea of using AI-driven visual content creation tools in their marketing tasks, whether they feel favorable or not. Having a positive attitude is quite important because it influences motivation and openness toward trying out new technologies, which in turn affects the intention to actually adopt them [53]. The importance of attitude in adopting technology has been noted in several previous studies. For example, [54] pointed out that attitude doesn't just impact someone's willingness to use new technologies but also plays a role in how other factors like PU and ease of use affect adoption. Studies specifically about AI adoption have shown that a positive attitude toward AI tools tends to increase the intention to use them [55] found that marketers who held a constructive attitude toward AI-enabled tools, including image generators, were more likely to bring these tools into their daily workflows. Similarly, [27] observed that attitude had a positive influence on the intention to adopt AI-based analytics within Bangladeshi SMEs. In Bangladesh's marketing environment, where AI technologies are still quite new, the attitudes of individuals towards AI image generators can either help or hinder their adoption. These attitudes are shaped by cultural background, how familiar users are with technology, and the perceived benefits that innovation might bring [38]. Also, having a positive attitude can help users overcome worries about risks and uncertainties related to AI-generated content, such as doubts about creativity, originality, or ethical concerns [56]. So, it can be said both theoretically and based on evidence that a favorable attitude towards AI image generators will probably lead to a stronger intention to use these tools. For organizations wanting to encourage AI adoption, understanding this relationship is key, and efforts to improve attitudes through awareness campaigns or training could really help boost acceptance rates. Even though prior literature has frequently treated Attitude as a bridging construct [24], other studies in technology adoption contexts have regarded it as a direct precursor of BI. This is because Attitude reflects users' judgmental and emotional stance toward the technology, which can directly drive the development of intention without always needing a mediator [57]. Considering the creativity- and feeling-driven nature of marketing practices, where professionals' enthusiasm or skepticism toward new tools exerts a direct effect on adoption, we define Attitude as an independent driver in our model. This specification is consistent with empirical findings in emerging technology adoption research that point to Attitude's direct effect on intention [19].

H5: ATT positively influences BI to use AI image generators.

PR generally refers to the possible negative outcomes or uncertainties that users associate with adopting and using a new technology [56,58]. When it comes to AI image generators, PR covers a range of concerns including the quality, reliability, originality, ethical issues, and even the potential misuse of AI-created content. Some of these worries might relate to fears of copyright violations, a lack of genuine human creativity, privacy problems, or losing control over how marketing messages are presented [59]. Many studies have shown that PR can act as a significant barrier when it comes to adopting new technologies. For instance, it has been pointed out by [60] that people's concerns about AI tools often reduce their willingness to adopt these technologies. Similarly, [61] emphasized that when individuals face uncertainty or potential negative consequences, their intention to use AI-based systems tends to decline. In marketing especially, where building and maintaining brand image and consumer trust is very important, the risks linked with AI-generated images become even more noticeable. Marketers might worry that AI visuals could be inaccurate, inappropriate, or might not genuinely connect with their audience [13]. Moreover, in countries like Bangladesh, where legal guidelines and public understanding about AI ethics are still developing, these risks might have an even stronger impact on whether marketers decide to adopt the technology or not [62]. Therefore, it can be expected that PR negatively affects the BI to use AI image generators. When marketers feel that risks are high, their hesitation to use these tools increases, which in turn slows down how quickly AI-powered creative technologies spread. It becomes essential, then, that these risks are recognized and addressed through clear communication, appropriate regulations, and user education to promote wider acceptance.

H6: PR negatively influences BI to use AI image generators.

PU is generally understood as how much a person believes that using a certain technology will improve their job performance or overall effectiveness [24]. While PE is quite similar conceptually and often considered the main predictor of BI in many technology acceptance theories, PU tends to represent a more focused cognitive judgment regarding the benefits of the technology as seen by its users [19,24,31]. While PE shows anticipations about efficiency benefits from AI image generators, PU denotes the user's mental evaluation of the technology's broad utility. Considering both constructs helps in covering task-focused anticipations and appraisal beliefs, thus offering a clearer understanding of BI development in marketing professionals. Because of this, PU can play a crucial mediating role in explaining why and how beliefs about performance translate into intentions to adopt AI image generators. Several studies have supported this mediating role of PU in different technology adoption contexts. For example, [63] extended the TAM by showing that PE influences BI indirectly through PU, suggesting that users tend to assess the usefulness of a technology before deciding to adopt it. In a similar vein, research by [38] found that PU acted as a mediator between PE and the intention to use AI-based analytics tools among SMEs in Bangladesh, emphasizing that perceptions of benefits drive adoption decisions. Specifically, when it comes to AI image generators, marketers' expectations that these tools will enhance their creativity and efficiency (PE) probably increase how useful they perceive these tools to be (PU), which in turn boosts their intention to use them (BI). This mediating effect seems particularly important in emerging markets like Bangladesh, where practical concerns and users' cognitive evaluations are shaped by factors such as resource limitations and varying digital literacy levels. Therefore, recognizing PU as a mediator helps in understanding the mental process through which performance-related beliefs are transformed into actual intentions to adopt technology, making it a key factor in explaining the adoption of AI image generators.

H7a: PU mediates the relationship between PE and BI to use AI image generators.

EE basically refers to how easy users believe technology is to use, and it plays a strong role in whether people are willing to adopt new systems [24]. However, the impact of EE on BI is not always direct. Instead, this relationship is often influenced by PU, which captures how much users think technology will help improve their job performance [64,65]. This means that when users find a system easy to operate, they tend to see it as more useful, and this perception of usefulness then encourages them to want to use it. Many studies have shown that the link between EE and BI is not always straightforward, with PU acting as a key cognitive bridge. For example, in their extension of the TAM2, [63] demonstrated that the ease of use affects perceived usefulness, which then influences BI. This mediating role of PU has also been found in various technology adoption contexts, including AI tools. Specifically, [38] reported that among SMEs in Bangladesh, PU mediated the relationship between EE and the intention to adopt AI-driven analytics, suggesting that when a technology is easier to use, people are more likely to find it useful and thus more willing to adopt it.

When it comes to AI image generators, marketing professionals are more likely to develop a positive perception of how useful the tool is if they feel it is easy to learn and fits smoothly into their work processes [66]. This is especially true in developing countries like Bangladesh, where users' digital literacy and available resources can vary a lot. Therefore, the ease with which they can interact with these AI tools greatly affects whether they recognize the practical value of such technology [67]. So, seeing PU as a mediator helps to explain the path through which EE shapes BI by influencing how users cognitively evaluate the benefits of AI image generators. For these reasons, it can be considered reasonable to propose that perceived usefulness mediates the effect of EE on BI to use AI image generators, shedding more light on the adoption process.

H7b: PU mediates the relationship between EE and BI to use AI image generators.

SI basically shows how much people feel that important others like their colleagues or managers think they should use a certain technology [31,43]. While SI can have a direct impact on someone's BI to adopt a technology, it often works indirectly through what's called PU. In other words, the social pressure or norms around a person tend to shape how useful they think a technology is, and that belief then influences whether they intend to use it or not. There have been several studies that point out this mediating role of PU between SI and BI [68] suggested that SI can change how users see the usefulness of technology, especially in workplaces or social groups. For example, if peers, supervisors, or influential people start using or recommending AI tools, individuals are more likely to think those tools are valuable and relevant for their own tasks. Supporting this, [38] showed with data from Bangladeshi SMEs that PU acts as a mediator for SI when it comes to adopting AI-driven analytics solutions. This means social approval increases how useful technology is perceived, which then encourages adoption. When it comes to AI image generators, marketers usually look at what their social circles think to judge the value of these new creative tools. In Bangladesh especially, where professional connections and peer suggestions hold a lot of weight, SI might boost how useful these marketers perceive AI image generators to be by confirming their practical benefits and easing doubts about the technology [13]. This helps turn social pressures into a mental assessment of usefulness, which ultimately shapes the intention to use technology. So, the mediating effect of PU is important to understand how SI leads to the intention to adopt technology. Seeing PU as a mediator offers a clearer picture of the mental process linking social expectations to actual BI in adopting AI technologies.

H7c: PU mediates the relationship between SI and BI to use AI image generators.

FC basically refers to how much people believe that their organization and technology setup provide the necessary support to use a certain technology [6]. While FC does have a direct effect on whether someone intends to use or uses technology, this influence is often channeled through PU. In other words, when users feel they have enough resources, assistance, and infrastructure, they tend to see the technology as more useful, which then encourages them to adopt it. Several previous studies have pointed out the role of PU as a mediator between FC and BI. For instance, [69] explained that FC can make users feel more confident and reduce obstacles, which then improves how useful they perceive the technology to be. Similarly, research by [38] showed that for SMEs in Bangladesh, PU acted as a bridge linking FC with the intention to adopt AI-powered analytics tools. This basically means that when support and infrastructure are in place, users are more likely to view the technology positively. Specifically, when it comes to AI image generators, marketing professionals tend to adopt these tools more readily if they believe their work environment provides enough support like training, IT help, and systems that work well with the technology [70]. Given the infrastructural challenges common in Bangladesh, these FC are even more important in shaping how useful marketers think these AI tools are. When marketers feel supported by their company and the tech environment around them, their confidence in the usefulness of AI image generators increases, which in turn raises their intention to use them [66]. So, understanding PU as a mediator between FC and BI helps explain how having the right support translates into people seeing the technology as useful, and thus being motivated to adopt it. This is especially true in developing countries, where infrastructure and support systems can vary a lot.

H7d: PU mediates the relationship between FC and BI to use AI image generators.

ATT basically reflects how positively or negatively an individual feels about using a particular technology [52]. While ATT has a direct effect on BI, this effect can be better understood when considering the mediating role of PU. In other words, when someone has a favorable attitude towards AI image generators, it often leads them to see the technology as more useful, which then makes them more likely to want to adopt it. Many studies have shown that PU plays a central role in technology acceptance models, acting as an important cognitive link between how people feel about technology (like their attitude) and their actual intentions to use it [63,71]. For instance, [60] pointed out that a person's attitude shapes how they perceive the benefits of a technology, and these perceptions influence their decisions to adopt it. Similarly, [38] found evidence from Bangladeshi SMEs that PU significantly mediates the relationship between attitude and intention to adopt AI-driven analytics tools, suggesting that attitude alone isn't enough unless it's paired with a belief that the technology is useful. Specifically, in marketing, especially with tools like AI image generators, professionals who feel positively about AI are generally more likely to appreciate how it can improve creativity and efficiency. This appreciation shows up as perceived usefulness, which is crucial in encouraging actual adoption. Moreover, in emerging economies such as Bangladesh, where acceptance of technology is influenced by both feelings and rational thoughts, considering PU as a mediator helps provide a clearer picture of how adoption happens [10]. So, it is important to examine PU as a mediator between ATT and BI, because this reveals how emotional attitudes translate into cognitive beliefs that drive the intention to use AI image generators.

H7e: PU mediates the relationship between ATT and BI to use AI image generators.

PR is often understood as the uncertainty and possible negative outcomes that users associate with adopting new technologies [58]. When PR is high, it tends to lower people's BI to use the technology, mainly because it shakes their trust and causes hesitation. However, PU can play an important mediating role in this relationship by shaping how users think about the benefits, even when risks are involved. The idea of PU as a mediator means that even if users see certain risks, if they believe the technology will be useful, they might still intend to adopt it [31,63]. In this sense, PU works like a cognitive filter, reducing the negative effect that risk perceptions might have [72] highlighted that PU could soften the impact of PR on technology acceptance, especially for AI-based tools where concerns about ethics and quality are often significant. In the field of marketing, AI image generators do come with risks like worries about originality of content, privacy of data, and ethical questions [38]. Still, when marketers see these tools as genuinely helpful for boosting creativity and efficiency, the negative influence of PR on their intention to adopt may be lessened. Supporting this, [27] found evidence from SMEs in Bangladesh showing that PU mediates the link between PR and BI, suggesting that a strong sense of usefulness can balance out the fears about new AI technologies. Considering that AI technologies are still new in Bangladesh's marketing sector, where users weigh both risks and benefits carefully, it becomes essential to understand PU as a mediator. This understanding helps explain how perceived usefulness can change risk perceptions into more positive intentions to adopt, which could guide organizations in crafting strategies that highlight usefulness while addressing risk concerns.

H7f: PU mediates the relationship between PR and BI to use AI image generators.

PU is generally seen as one of the most important factors influencing whether users decide to adopt new technologies [24,31]. It basically refers to how much a person believes that using a certain system or tool will help improve their job performance or overall effectiveness. When it comes to AI image generators, PU reflects marketers' thoughts on how these tools might boost their creative output, make content creation easier, and eventually lead to more successful marketing campaigns. Many studies have shown that PU plays a key role in shaping BI across a range of technologies. For instance, [63] found that PU strongly and directly affects BI in their updated TAM2. Similar findings have been reported in studies related to AI adoption [38], for example, observed that marketers' views on the usefulness of AI-driven image generation had a significant impact on their willingness to use these tools for digital marketing. Specifically, in the Bangladeshi context where limited resources and intense competition are common the perceived benefits of AI image generators in improving marketing efficiency and creativity become especially important in deciding whether to adopt the technology [38,27] also showed through their research with Bangladeshi SMEs that PU was a strong predictor of BI to use AI-powered analytics, highlighting its critical role in emerging economies. Considering how central PU is in connecting different factors of adoption and its direct influence on BI, it becomes quite necessary to clearly investigate its positive effect on BI to use AI image generators. Understanding this relationship allows organizations and marketers to focus on enhancing perceptions of usefulness, which in turn can promote wider acceptance and use of such technologies.

H8: PU positively influences BI to use AI image generators.

Based on the hypothesized relationships derived from the reviewed literature and theoretical arguments, the conceptual framework for this study has been meticulously developed. This framework, illustrated in Fig. 1, encapsulates the key constructs and their interrelationships, offering a visual representation of the study's underlying theoretical foundations. Each element within the framework is strategically positioned to reflect the logical flow and connectivity between the variables,

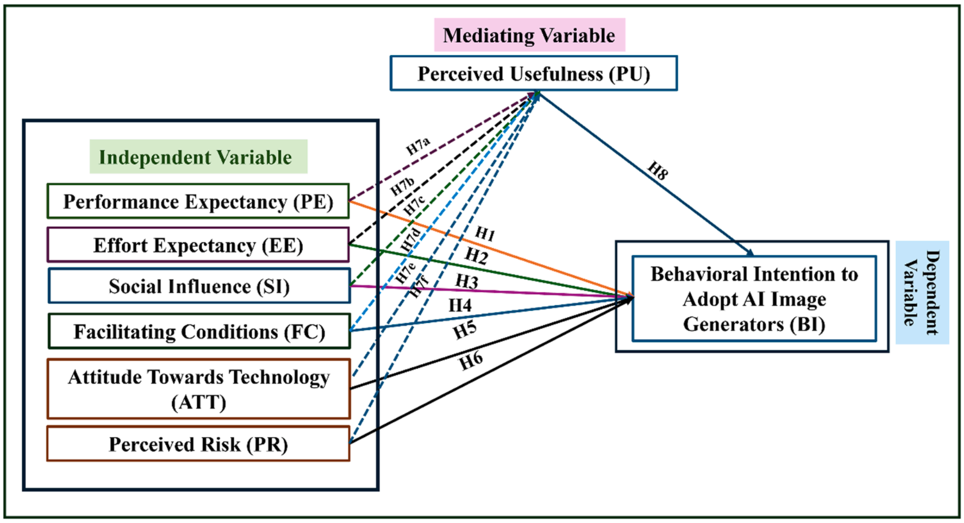

Fig. 1. Conceptual framework.

providing a comprehensive overview of the anticipated dynamics explored in the research. Fig. 1 serves as a crucial guide, bridging the gap between theoretical postulations and empirical investigation.

3. Materials and method

This study was conducted using a quantitative research approach to explore the BI of marketing professionals in Bangladesh toward adopting AI image generators. The quantitative method was chosen mainly because it allows for measurable, structured data that can be statistically analyzed [73,74]. This approach seemed especially suitable for testing the hypothesized relationships among key factors such as PE, EE, SI, FC, ATT, PR, PU, and BI within a well-established theoretical framework. Collecting primary data was important here to gain specific and timely insights from real professionals actively involved in marketing.

The population targeted in this research consisted of marketing professionals working in various sectors across Bangladesh. These individuals were selected because they are more likely to be exposed to digital tools and AI applications in their promotional campaigns. Since AI image generators are still relatively new in the local context, marketing professionals were seen as appropriate respondents due to their role in strategy formulation, campaign execution, and the adoption of new technologies. The unit of analysis was the individual professional rather than the organization, focusing on personal perceptions, attitudes, and intentions. Because there is no comprehensive database or official list of marketing professionals in Bangladesh, this study made use of convenience sampling to select respondents. In countries like Bangladesh where professional directories are limited or unavailable, convenience sampling often becomes a practical choice [75]. While this method has its drawbacks, especially regarding selection bias, steps were taken to reduce such limitations. For example, the study ensured respondent diversity by contacting professionals from different types of organizations ranging from SMEs to large firms and advertising agencies across various sectors and regions. Questionnaires were distributed both face-to-face and online via email, LinkedIn, and professional networks to increase reach. Screening questions helped include only those who had relevant experience or were involved in marketing technology adoption decisions. These measures aimed to reduce the risk of bias while still allowing efficient data collection.

A total of 480 questionnaires were distributed, from which 343 responses were returned, indicating a decent response rate. After cleaning the data to remove incomplete or inconsistent answers, 320 valid responses remained for analysis. This sample size was not chosen randomly but based on a power analysis using G*Power software following [76,77] recommendations. With a medium effect size ($f^2 = 0.15$), 80% power, a 5% significance level, and up to five predictors per construct, the minimum sample size required was about 107. Therefore, the final sample of 320 was more than adequate to provide reliable and statistically meaningful results.

The questionnaire items were adapted from validated scales in previous studies and measured on a five-point Likert scale ranging from 1 ("Strongly Disagree") to 5 ("Strongly Agree"). This scale was chosen for its simplicity and ease of use, which is important in developing countries where more complex rating systems might confuse respondents or reduce response quality [78]. The questions were slightly modified to fit the specific context of AI image generator adoption.

Before the main survey, a pilot test was conducted with a small group of 15 marketing professionals and academic experts to check the clarity and relevance of the questionnaire. Based on their feedback, some items were rephrased and adjusted to improve understanding and reduce ambiguity, which improved the reliability of the instrument. To ensure content and construct validity, experts reviewed the survey, and during analysis, convergent and discriminant validity were assessed using indicator loadings, Average Variance Extracted (AVE), Composite Reliability (CR), and the Fornell-Larcker criterion with SmartPLS 4.0 software.

Data entry and initial cleaning were done using SPSS, while the main data analysis was performed using SmartPLS 4.0. This software was selected because it can effectively handle complex models involving multiple indicators and mediating effects, even when data does not follow a normal distribution. The use of PLS-SEM was justified for several reasons. Primarily, the study focused more on prediction and theory development rather than strict theory testing, which aligns with PLS-SEM strengths [79]. The model also included mediating relationships, and several latent variables measured reflectively, conditions under which PLS-SEM works well. Additionally, PLS-SEM is known to be robust with relatively smaller sample sizes and data that do not meet multivariate normality assumptions, fitting the characteristics of this research dataset. Moreover, SmartPLS provided powerful bootstrapping tools to test the significance of direct, indirect, and mediating effects, which was particularly important for evaluating the mediating role of PU. The software also supported a combined assessment of the measurement model checking for reliability and validity and the structural model, including path coefficients and overall model fit. This enabled a thorough and integrated analysis of the data collected.

The reliability coefficients for each construct involved in the study are shown in Table 1. The measurement constructs of EE and PR demonstrated strong reliability, as shown by Cronbach's alpha

Table 1
Reliability of the measurements.

Constructs	Item	Cronbach's alpha
Performance Expectancy	3	0.91
Effort Expectancy	3	0.92
Social Influence	3	0.80
Facilitating Conditions	3	0.71
Attitude Towards Technology	3	0.72
Perceived Risk	3	0.90
Perceived Usefulness	3	0.71
Behavioral Intention to adopt AI Image Generators	3	0.77

coefficients of 0.92 and 0.90, respectively. The presence of a robust internal consistency implies that the items within these constructs consistently and accurately assess their intended constructions [80]. Furthermore, the construct of PE exhibited strong internal consistency, as seen by its excellent reliability with a Cronbach's alpha of 0.91. The Cronbach's alpha coefficient for BI to Adopt AI Image Generators was 0.77, indicating a satisfactory degree of reliability for assessing the intention to utilize AI image generators [79]. SI presented Cronbach's alpha coefficient of 0.80, indicating a dependable assessment of the impact of social elements on BI. In contrast, the Constructs of ATT, FC, and PU received Cronbach's alpha coefficients of 0.72, 0.71, and 0.71, correspondingly. Although these results fall inside the lower limit of the permitted range, they nonetheless demonstrate enough dependability for exploratory study. The appropriate range for Cronbach's alpha is often between 0.70 and 0.90, depending upon the specific circumstances and the phase of the study. In general, the reliability analysis verifies that the measurement scales used in this work exhibit suitable consistency, therefore enabling a dependable evaluation of the constructs associated with the deployment of AI picture generators in marketing.

4. Results

The demographic profile of the respondents in the research is shown in Table 2, offering valuable insights into the general makeup of the sample. With respect to gender, the sample comprises 64.0% men (N=206) and 36.0% females (N=114), suggesting a greater proportion of males in the group of respondents. The observed gender distribution is consistent with industry standards, where male professionals tend to be more prevalent in certain areas (Smith & Duggan, 2013). Among the responders, a significant proportion (74.7%, N=239) are between the ages of 27 and 42 years. This demography suggests that the survey mostly included the viewpoints of professionals in the middle stage of their careers, who are expected to possess significant expertise and impact in marketing choices.

In contrast, the younger age group (15.0%, N=48) and the older age group (10.3%, N=33) are under-represented, perhaps reflecting the contemporary distribution of the marketing industry personnel. Analysis of current job status reveals that the majority of respondents are mid-level employees (70.3%, N=225), indicating a preference for those who have significant industry experience and responsibility. Entry-level personnel account for 16.9% (N=54), while senior-level employees make up 12.8% (N=41), indicating a mostly mid-level responder group. In terms of marketing experience, a substantial bulk of respondents (61.9%, N=198) had 4-6 years of experience, suggesting a notable percentage of participants with intermediate experience. The group with 1-3 years of experience accounts for 19.1% (N=61), while those with 7-10 years (13.8%, N=44) and above 10 years (5.3%, N=17) are less prevalent. This distribution indicates that the survey records the perspectives of professionals who have accumulated a considerable amount of experience but have not yet reached the most advanced phases of their professions. The breakdown of organization types reveals that 38.4% of the participants are employed in marketing agencies (N=123) and 35.6% are employed in advertising agencies (N=114). These findings suggest a substantial presence from both the specialized marketing and advertising industries. The proportion of freelancers is 18.4% (N=59), whilst in-house marketing departments account for 3.4% (N=11) and other sorts of organizations make up 4.1% (N=13). This distribution represents the broad range of situations in which marketing professionals work, therefore providing a wide range of viewpoints to the research.

The measurement model findings for the constructs employed in the investigation are shown in Table 3, offering valuable insights into their reliability and validity. Item loadings, Average Variance Extracted (AVE), and Composite Reliability (CR) for each construct are provided in the table. PE has strong reliability, as shown by item loadings between 0.88 and 0.94, an AVE of 0.85, and a CR of 0.94. The obtained results demonstrate robust convergent validity and reliability, as all loadings above the criterion of 0.70, with both AVE and CR exceeding the acceptable criteria of 0.50 and 0.70, respectively [79]. The observation implies that the items successfully measure the concept of PE and

Table 2
Demographic profile of the respondents.

Variable	N	(%)
Gender		
Male	206	64.0
Female	114	36.0
Age		
26-35 Years	48	15.0
36–45 Years	239	74.7
46–58 Years	33	10.3
Current Employment Status		
Entry-level employee	54	16.9
Mid-level employee	225	70.3
Senior-level employee	41	12.8
Years of Experience in Marketing		
1-3 years	61	19.1
4-6 years	198	61.9
7-10 years	44	13.8
More than 10 years	17	5.3
Organization Type		
Marketing agency	123	38.4
Advertising agency	114	35.6
In-house marketing department	11	3.4
Freelancer	59	18.4
Others	13	4.1
Total	320	100%

Table 3
Measurement model.

Constructs	Items	Loading	AVE	CR
Performance Expectancy	PE1	0.88	0.85	0.94
	PE2	0.93		
	PE3	0.94		
Effort Expectancy	EE1	0.95	0.86	0.95
	EE2	0.93		
	EE3	0.91		
Social Influence	SI1	0.74	0.72	0.88
	SI2	0.88		
	SI3	0.92		
Facilitating Conditions	FC1	0.87	0.64	0.84
	FC2	0.63		
	FC3	0.88		
Attitude Towards Technology	ATT1	0.77	0.64	0.84
	ATT2	0.86		
	ATT3	0.76		
Perceived Risk	PR1	0.92	0.83	0.93
	PR2	0.92		
	PR3	0.89		
Perceived Usefulness	PU1	0.82	0.64	0.84
	PU2	0.87		
	PU3	0.69		
Behavioral Intention to adopt AI Image Generators	BI1	0.85	0.69	0.87
	BI2	0.88		
	BI3	0.75		

enhance their strong dependability. EE demonstrates strong empirical evidence with item loadings ranging from 0.91 to 0.95, AVE of 0.86, and CR of 0.95. The metrics provide as evidence of high reliability and convergent validity, therefore validating the consistent and accurate measurement of the concept [77]. The item loadings for SI range from 0.74 to 0.92, with an AVE of 0.72 and a CR of 0.88. Although the loadings and CR are acceptable, the AVE is rather higher than the minimal requirement of 0.50, indicating sufficient CV [77,79]. The FC exhibit item loadings ranging from 0.63 to 0.88, an AVE of 0.64, and a CR of 0.84. The AVE value above the criterion of 0.50.

However, the lower loading of FC2 (0.63) may rather impact the dependability of the construct. Nevertheless, the build is typically dependable, but further modifications may be necessary. The item loadings for ATT range from 0.76 to 0.86, with an AVE of 0.64 and a CR of 0.84. The results indicate strong convergent validity and reliability. However, the performance of the construct might be further improved by increasing the item loading. The PR variable has item loadings ranging from 0.89 to 0.92, an AVE of 0.83, and a CR of 0.93. These results demonstrate strong convergent validity and reliability, as all loadings and the AVE above the suggested criteria [79]. The item loadings for PU range from 0.69 to 0.87, with an AVE of 0.64 and a confidence ratio (CR) of 0.84. The loading of PU3 (0.69) falls somewhat below the optimal threshold of 0.70, potentially impacting the dependability of the construct to some extent [79]. Nevertheless, the general framework remains sufficient for the objectives of the research. The BI to Adopt AI Image Generators exhibits item loadings ranging from 0.75 to 0.88, an AVE of 0.69, and CR of 0.87. The observed results indicate strong convergent validity and reliability, which comply with the established criteria for accurate measurement as outlined by Nunnally and Bernstein [81]. The measurement model exhibits robust reliability and validity for most constructs, while there are some aspects that might be improved to increase measurement accuracy. The results validate the reliability of the constructs in assessing the acceptance of AI image generators in marketing.

Table 4 presents the results of the 'Fornell-Larcker criterion, which is used to assess the discriminant validity of the constructs in the study. The Fornell-Larcker criteria evaluates the correlations across constructs by computing the square root of the AVE for each construct. The diagonal numerical values in Table 4 correspond to the SR of the AVE for each construct. In order to show discriminant validity, these values should exceed the off-diagonal correlations [79,82]. The SR of the 'AVE for ATT is 0.80, which is greater than its correlations with other constructs including BI (0.59), EE (0.56), FC (0.68), PR (0.57), PU (0.55), PE (0.56), and SI (0.55). BI has a SR of 'AVE of 0.83, which indicates that its value is higher than the correlations with ATT (0.59), EE (0.59), FC (0.57), PR (0.61), PU (0.67), PE (0.67), and SI (0.82). The evidence indicates that BI is well differentiated from the other constructs, therefore validating its discriminant validity [79]. The SR of 'AVE' for EE is 0.93, which is significantly greater than its correlations with ATT (0.56), BI (0.59), FC (0.62), PR (0.54), PU (0.87), PE (0.72), and SI (0.63). This large number suggests a high level of discriminant validity for EE. A SR of the 'AVE' for FC is 0.80. By virtue of its greater connections with ATT (0.68), BI (0.57), EE (0.62), PR (0.84), PU (0.60), PE (0.86), and SI (0.70), FC is clearly distinguishable from the other constructs. The SR of 'AVE' for PR is 0.91, which exceeds its relationships with ATT (0.57), BI (0.61), EE (0.54), FC (0.84), PU (0.57), PE (0.86), and SI (0.76). This illustrates that PR has strong discriminant validity. The SR of 'AVE' for PU is 0.92, which is higher than its average correlations with ATT (0.55), BI (0.67), EE (0.87), FC (0.60), PR (0.57), PE (0.72), and SI (0.65). These findings validate the uniqueness and sufficient discriminant validity of the concept [79,82]. Based on its square root of AVE of 0.85, PE has stronger connections with ATT (0.56), BI (0.67), EE (0.72), FC (0.86), PR (0.86), PU (0.72), and SI (0.77), suggesting that PE is unique among the constructs. The SR of AVE for SI is 0.80, surpassing its associations with ATT (0.55), BI (0.82), EE (0.63), FC (0.70), PR (0.76), PU (0.65), and PE (0.77). Table 4 illustrates that all constructs in the investigation show sufficient DV according to the FLC. The SR of the AVE for each construct exceeds its 'correlations with other constructs, therefore confirming the uniqueness of each construct in the measurement model.

Table 4
Fornell Larcker criterion among constructs.

Constructs	ATT	BI	EE	FC	PR	PU	PE	SI
ATT	0.80							
BI	0.59	0.83						
EE	0.56	0.59	0.93					
FC	0.68	0.57	0.62	0.80				
PR	0.57	0.61	0.54	0.84	0.91			
PU	0.55	0.67	0.87	0.60	0.57	0.92		
PE	0.56	0.67	0.72	0.86	0.86	0.72	0.85	
SI	0.55	0.82	0.63	0.70	0.76	0.65	0.77	0.80

When it comes to Variance Inflation Factor (VIF) values, the accepted rule of thumb is that values below 5 are generally considered safe, meaning multicollinearity among the predictor variables is not a serious issue [77,83]. Looking at Table 5, all the VIF values for both direct and mediated paths stayed under this cutoff, which suggests that multicollinearity did not significantly affect the regression estimates in this study. More specifically, the VIF values for the direct predictors of BI such as PE with 4.68, EE at 4.88, SI 3.54, FC 2.43, ATT 2.15, and PR 4.96 all fell within the acceptable limits. This shows that although there may be some correlation between these predictors, it wasn't strong enough to cause problematic multicollinearity that would interfere with interpreting the relationships between variables.

Additionally, when considering the mediating effects through PU, the VIF values also remained well below 5. For instance, in the path from PE to PU to BI, PE had a VIF of 4.68 and PU had 2.96, indicating a good level of independence among these constructs. The same pattern was seen in other mediation paths EE, SI, FC, ATT, and PR each paired with PU maintained VIF values comfortably under the threshold. This consistency strengthens the overall structural soundness of the model and supports the use of PLS-SEM, which is well-known to handle complex models with mediators and reflective indicators effectively [77,79].

Table 6 presents the results of the hypothesis testing for the study on AI image generator adoption in marketing within the Bangladeshi context. The findings reveal several significant relationships between the constructs, confirming the mediating role of PU in certain cases. H1, which posits that PE positively influences BI to use AI image generators, was supported with a T-statistic of 4.22 and a p-value of 0.00, indicating a strong and significant relationship. This finding aligns with previous studies that have demonstrated the importance of PE in technology adoption decisions, particularly in contexts where the perceived benefits of the technology directly impact user intentions [38]. H2, H3, H4, and H5, which explore the relationships between EE, SI, FC, and ATT with BI, were all accepted, with p-values of 0.00. These results underscore the significance of ease of use, social norms, supportive conditions, and positive attitudes in fostering the intention to adopt AI technologies in

Table 5
Inner model VIF values for predictor constructs.

Constructs	Inner VIF Values	VIF < 5
PE → BI	4.68	✓
EE → BI	4.88	✓
SI → BI	3.54	✓
FC → BI	2.43	✓
ATT → BI	2.15	✓
PR → BI	4.96	✓
PE → PU → BI	4.68 (PE), 2.96 (PU)	✓
EE → PU → BI	4.88 (EE), 2.96 (PU)	✓
SI → PU → BI	3.54 (SI), 2.96 (PU)	✓
FC → PU → BI	2.43 (FC), 2.96 (PU)	✓
ATT → PU → BI	2.15 (ATT), 2.96 (PU)	✓
PR → PU → BI	4.96 (PR), 2.96 (PU) 2.96	✓
PU → BI	2.96	✓

Table 6
Hypothesis testing.

Hypothesis	Relationship	T statistics	P values	Decision
H1	PE → BI	4.22	0.00	Accepted
H2	EE → BI	3.01	0.00	Accepted
H3	SI → BI	3.09	0.00	Accepted
H4	FC → BI	3.83	0.00	Accepted
H5	ATT → BI	3.56	0.00	Accepted
H6	PR → BI	1.74	0.08	Rejected
H7a	PE → PU → BI	2.03	0.04	Accepted
H7b	EE → PU → BI	1.35	0.18	Rejected
H7c	SI → PU → BI	2.22	0.03	Accepted
H7d	FC → PU → BI	0.83	0.41	Rejected
H7e	ATT → PU → BI	1.75	0.08	Rejected
H7f	PR → PU → BI	5.27	0.00	Accepted
H8	PU → BI	13.89	0.00	Accepted

marketing. However, PR did not significantly influence BI, as indicated by the rejection of H6 (p-value of 0.08). This suggests that while PR may be a concern, they do not outweigh the perceived benefits or other facilitating factors when it comes to the decision to adopt AI image generators. This is an interesting finding, as it contrasts with some studies that emphasize the role of PR as a barrier to technology adoption, especially in emerging markets. The mediating role of PU was confirmed for certain relationships. Hypotheses H7a (PE → PU → BI), H7c (SI → PU → BI), and H7f (PR → PU → BI) were accepted, indicating that PU significantly mediates the relationship between these factors and BI. This highlights the critical role of PU in enhancing the adoption of AI image generators, as users are more likely to adopt the technology if they find it useful in achieving their marketing goals. Conversely, H7b (EE → PU → BI), H7d (FC → PU → BI), and H7e (ATT → PU → BI) were rejected, suggesting that PU does not mediate the relationship between EE, FC, ATT, and BI in this context. These findings may indicate that while these factors are important, their influence on adoption intention is more direct and less reliant on the PU of the technology. Finally, PU itself was found to have a strong and significant direct effect on BI (H8), with a T-statistic of 13.89 and a p-value of 0.00. This confirms the pivotal role of PU in driving the adoption of AI image generators in marketing, echoing the findings of numerous studies that position PU as a central determinant of technology adoption.

The R^2 and R^2 adjusted values for the constructs in this study provide valuable information about the statistical ability of the research model to make accurate predictions. A coefficient of determination (R^2) of 0.75 for BI to adopt AI Image Generators suggests that 75% of the variability in BI can be accounted for by the independent variables in the model. These variables include PE, EE, SI, FC, ATT, and PR. A high R^2 value indicates a robust explanatory capability of the model, underscoring the significance of these aspects in forecasting the acceptance of AI image generators in the marketing environment of Bangladesh [83]. The coefficient of determination (R^2) for PU is 0.66, indicating that 66% of the variability in PU can be explained by its antecedents, including PE, EE, SI, enabling factors, and ATT. Thus, these elements together have a substantial influence on consumers' perception of the utility of AI image generators, which is essential for their acceptance in marketing strategies [77,84]. The modified R^2 values, 0.74 for BI and 0.66 for PU, are somewhat lower than the R^2 metrics but maintain their robustness. The small discrepancy between the R^2 and R^2 adjusted values indicates that the model is well defined and adequately fits the data. Adding more predictors is unlikely to greatly enhance the model's ability to explain the data. The consistency seen in this research serves to strengthen the validity of the constructs and the strength of the associations that were established. The R^2 and R^2 adjusted values indicate that the theoretical model well captures the main factors influencing the adoption of AI image generators in marketing in the specific context of Bangladesh. The results emphasize the need to prioritize these specified elements to increase the rates of adoption, especially in areas where the incorporation of technology in marketing is still in its preliminary stages. (Table 7)

Table 7
R^2 and R^2 adjusted.

Constructs	R^2	R^2 adjusted
Behavioral Intention to adopt AI Image Generators	0.75	0.74
Perceived Usefulness	0.66	0.66

The SEM diagram depicts the interconnections among several factors and their influence on the BI to Use AI Image Generators and the Perceived Behavioral Usefulness. The route coefficients, shown by the arrows connecting the constructs, indicate the significance and orientation of these associations. The mediating function of PU is of utmost importance in this model, since it has a substantial direct influence on BI (path coefficient = 0.70). This conclusion aligns with the evidence of [31] that PU is a robust predictor of technology uptake. The path coefficient of 0.44 indicates that FC have the greatest influence on PU. This implies that the presence of resources and support systems greatly improves users' opinion of the usefulness of AI image generators. This underscores the need of guaranteeing that consumers are provided with the essential tools and support to efficiently use AI technology. In addition, EE has a positive impact on PU (path coefficient = 0.28), suggesting that people are more inclined to see the technology as valuable if it is user-friendly. PU is positively influenced by ATT (path coefficient = 0.27), indicating that a good ATT in general might increase the perceived advantages of AI applications. Conversely, the negative influence of PR on PU (path coefficient = -0.26) and BI (path coefficient = -0.17) highlights the need of addressing the dangers and uncertainties related to AI technology to enhance its adoption. The correlation coefficients (-0.19 and -0.07, respectively) between SI and both PU and BI are rather weak and non-significant. This implies that social determinants may not have a significant impact on shaping perceptions or intentions regarding AI image generators in this setting. With R-square values of 0.75 for BI and 0.66 for PU, the model exhibits a robust overall fit, suggesting that it accounts for a significant amount of variation in both dimensions. The results align with the UTAUT paradigm, which suggests that the perceived efficiency and enabling circumstances play a crucial role in the adoption of technology [24,31]. (Fig. 2)

5. Discussion

This research set out to explore the different behavioral, technological, and contextual elements that shape the adoption of AI image generators among marketing professionals in Bangladesh. While the global marketing landscape is increasingly leaning into AI-driven visuals [24], the pace of adoption in developing countries like Bangladesh has been more uneven largely due to limitations in digital infrastructure, varying cultural perceptions, and the level of institutional support available [13,27]. Drawing from the original UTAUT framework and extending it with ATT and PR, this study took a closer look at both the psychological and environmental triggers that influence a marketer's decision to adopt such tools [31]. A particular focus was placed on the mediating role played by PU, and whether it helps to explain the link between constructs like PE, EE, SI, FC, ATT, and PR on the intention to adopt (BI) these technologies.

From the findings, it became clear that constructs such as PE, EE, SI, FC, and ATT have a strong and direct influence on BI, echoing earlier claims that users are more inclined to adopt technologies they find useful, user-friendly, socially validated, and supported by the right conditions [31,38,79]. Curiously, PR was not found to have a significant impact on BI. This could indicate that marketers in Bangladesh aren't particularly deterred by the potential downsides of AI adoption be it ethical dilemmas, job threats, or data-related concerns possibly due to a general sense of optimism around digital tools or limited awareness regarding such risks [10,27]. In addition, the study confirms that PU does mediate several key relationships especially those involving PE, SI, and PR signaling that the PU of the tool partly explains why these

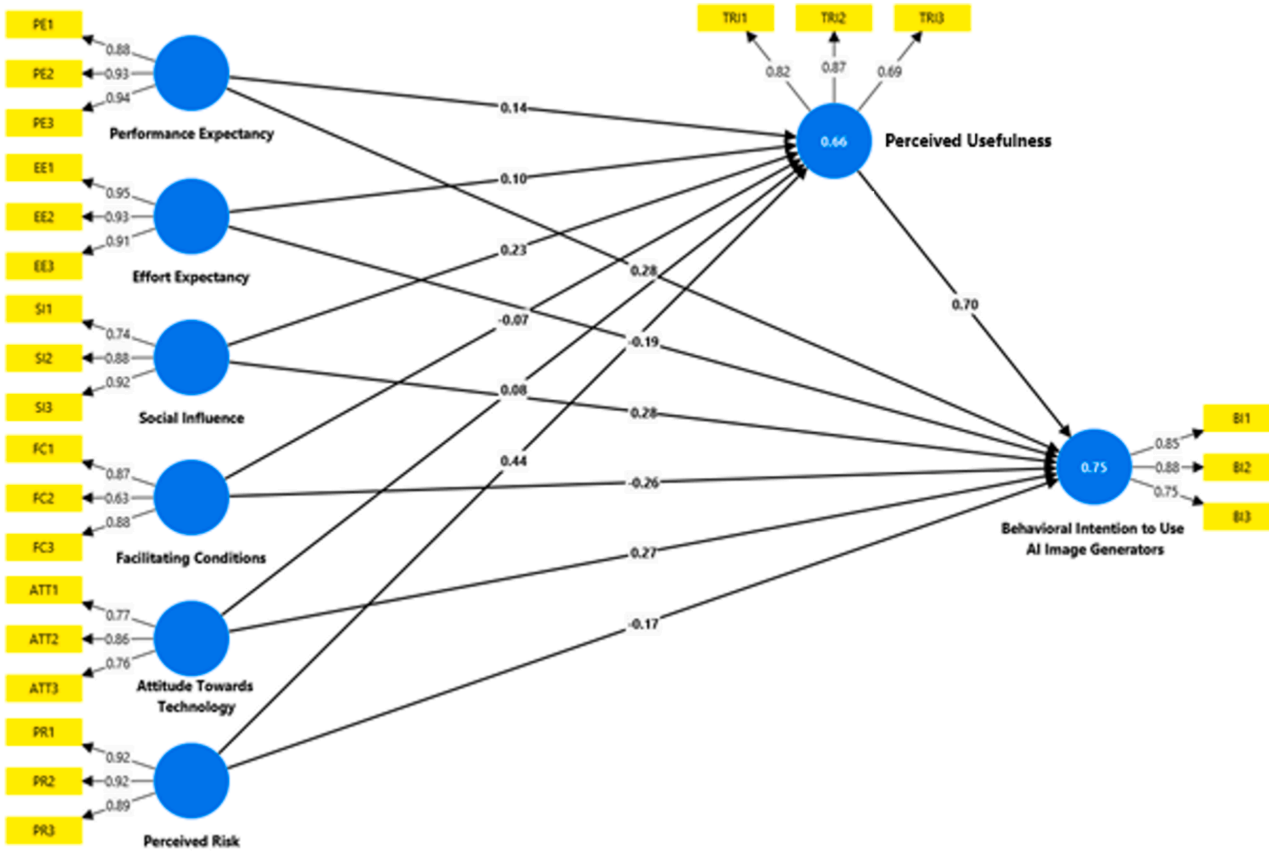

Fig. 2. Structural equation modelling (PLS-SEM approach).

variables matter to adoption intention [24].

However, PU didn't serve as a mediator for EE, FC, and ATT, suggesting that these factors influence BI in a more direct manner. For example, when a system is considered easy to use or there are proper facilities in place, users might be encouraged to adopt it regardless of how useful they find it [38,59]. These subtle distinctions enrich the broader understanding of the adoption process and point to the fact that not all motivators work through the same cognitive lens.

Theoretical & managerial implications

From a theoretical perspective, this study enhances the classic UTAUT framework by tailoring it to better fit the realities of developing markets and newer tech contexts. By bringing in ATT and PR and closely examining the role of PU as a bridge between variables and intention, the study introduces a more layered and inclusive model that captures how marketers approach AI image tools in their work [13,31,77]. It suggests that PU is more than just a direct driver. It acts as a vital link connecting behavior, perception, and environment [59,79]. These insights make the framework more adaptable for studying adoption behavior in creative fields, where AI tools are reshaping workflows and expectations [13].

On the managerial side, these results offer some practical insights for developers, marketers, and policy actors. Since PE and PU turned out to be strong predictors of adoption, developers should highlight specific advantages like improved content quality, creative freedom, or time efficiency in their marketing strategies [39]. At the same time, simplifying the user experience and providing proper onboarding can reduce EE, making it easier for users to feel confident with the tool [66]. With SI playing a strong role, organizations could benefit from promoting peer success stories, expert endorsements, or industry influencer campaigns to normalize adoption within marketing teams [31,39]. And because PR wasn't a significant barrier, this could allow firms to allocate fewer resources to risk communication and instead focus on usability, performance, and user support [41].

6. Conclusion

This study set out with the intention to better understand what really drives the use of AI image generators in marketing, particularly in the Bangladeshi context. While AI tools have been gaining traction globally, there hasn't been much exploration of how they're being adopted in developing countries, especially in the marketing space. To help fill that gap, this study investigated how individual perceptions like users' attitudes and worries as well as external factors such as the availability of support or the simplicity of using these tools, influence adoption in a setting where AI is still relatively new. The research extended the well-known UTAUT model by adding two more elements: ATT and PR. These additions helped paint a more complete picture. Results showed that both FC and EE play a meaningful role in shaping PU, which then strongly affects BI to adopt AI image generation tools. Interestingly, PR turned out to be a barrier, which underlines how important it is to build trust and reduce uncertainty around these technologies. The model worked well overall, explaining about 75% of the variance in BI and 66% in PU solid numbers that suggest the framework captured much of the adoption process in this context. From a theoretical angle, adding ATT and PR to the model contributes to a richer understanding of how tech is accepted in developing economies. In practical terms, the findings offer valuable guidance to marketers, tech designers, and policymakers who want to make AI tools more appealing. By focusing on ease of use, building digital support systems, and tackling the fears people might have about AI, adoption could be boosted significantly. In the long run, this could help create a more forward-thinking and competitive marketing environment in Bangladesh, where professionals feel empowered to use AI to improve both creativity and efficiency.

Limitations

This study is not without limitations. For one, it was only carried out in Bangladesh, so the findings might not translate perfectly to other countries where tech access, education levels, or even cultural attitudes could be quite different. Another thing to keep in mind is the cross-sectional nature of the research; it only gives us a snapshot in time. People's opinions and behaviors could shift as AI tools become more common, and this study wasn't able to capture those changes in this study.

Future research directions

Looking ahead, future studies could explore AI adoption in other industries or regions to see how the results compare. That would help confirm whether these findings hold true elsewhere or are unique to the Bangladeshi context. Longitudinal research would also be useful, as it could track how people's attitudes and behaviors change over time as they get more familiar with AI tools. It might also be worth digging deeper into how cultural values and norms affect people's openness to AI. Doing so could help create adoption models that are more sensitive to the realities and mindsets of different communities around the world.

CRediT authorship contribution statement

Md Mehedi Hasan Emon: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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